

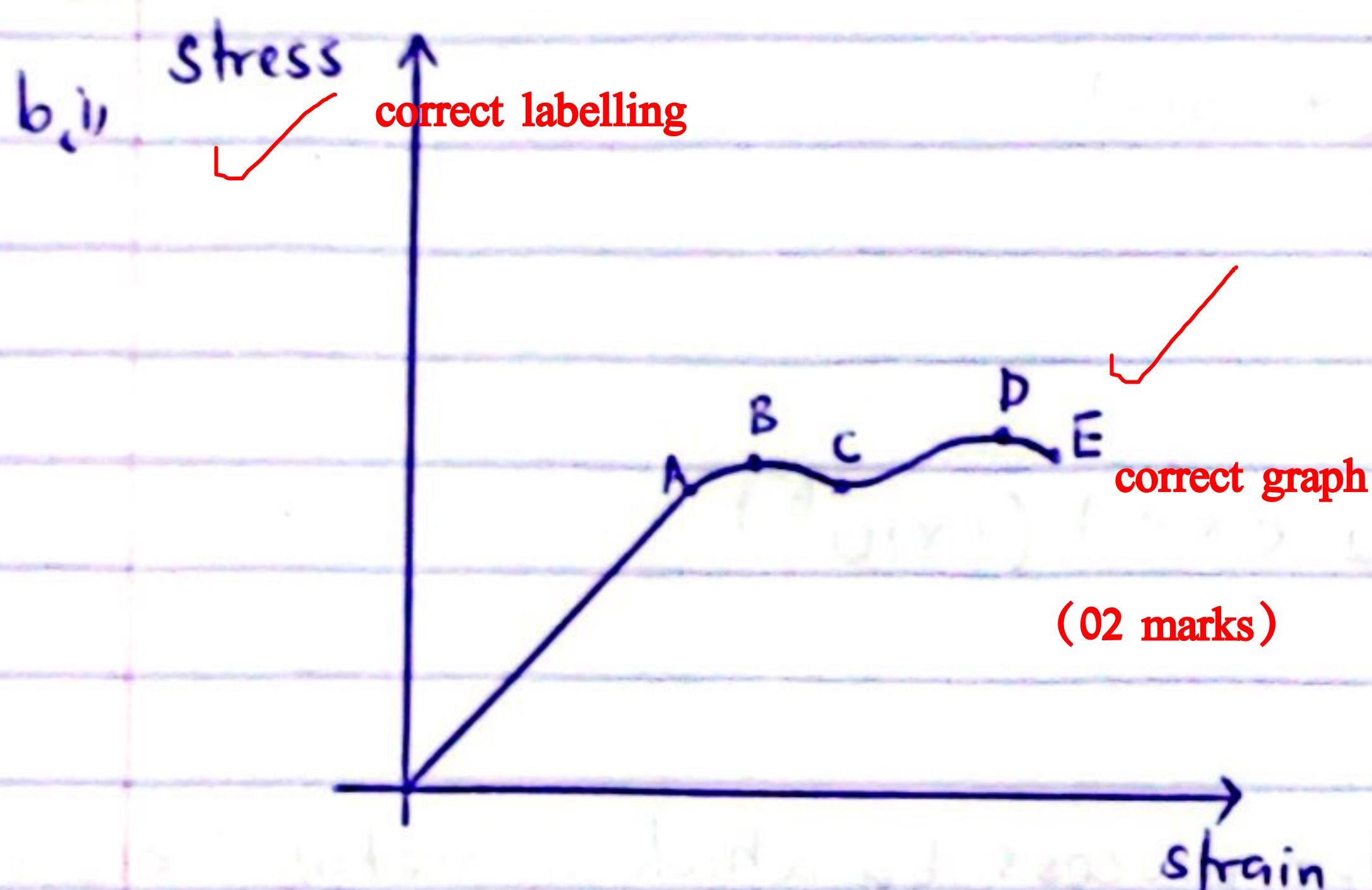
S-6 TEST ONE GUIDE, 2026



1a) Tensile stress is the pulling force per unit cross sectional area applied to a material. ✓ (01 marks)

ii) Tensile strain is fractional change in length of a material when stretched.

It is the ratio of extension to original length. ✓ (01 marks)



- A - Proportional limit
- B - Elastic limit
- C - Yield point
- D - Breaking stress
- E - Breaking point

OA - Strain is directly proportional to stress. The material obeys Hooke's law. ✓

AB - Stress is not proportional to strain but the material regains its length. ✓

Beyond B - the material becomes permanently stretched.

CD - the material undergoes plastic deformation. ✓

Beyond D - the material breaks. ✓

(04 marks)

ii) In elastic region, energy supplied is stored in the wire as elastic potential energy (strain energy) ✓

Between elastic and yield point, part of energy is stored as elastic energy and some used in rearranging the internal structure of the wire. ✓

Beyond yield point, energy is used in overcoming internal forces between atoms, causing layers of atoms to slide past each other. ✓

Near breaking point, energy is released as heat, sound and l.e of the broken ends. ✓ (04 marks)

c) i) For a small extension, dx

Work done, $dw = F dx$ ✓

But $F = kx$

$W = \int dw$

$= \int_0^e kx dx$ ✓

$$= \frac{1}{2} k e^2$$

(03 marks)

ii) $e = \alpha L \Delta \theta$

$$\frac{e}{L} = \alpha \Delta \theta$$

Thermal strain = $\frac{e}{L} = \alpha \Delta \theta$

$$= (1.1 \times 10^{-5}) (40 - 20)$$

$$= 2.2 \times 10^{-4}$$

(03 marks)

iii) $F = \alpha \Delta \theta E A$

$$= (1.1 \times 10^{-5}) (40 - 20) (2.0 \times 10^{-4}) (1 \times 10^{-6})$$

$$= 44 \text{ N}$$

(03 marks)

(d) Work hardening is the process by which a metal becomes stronger and harder when it is plastically deformed

importances

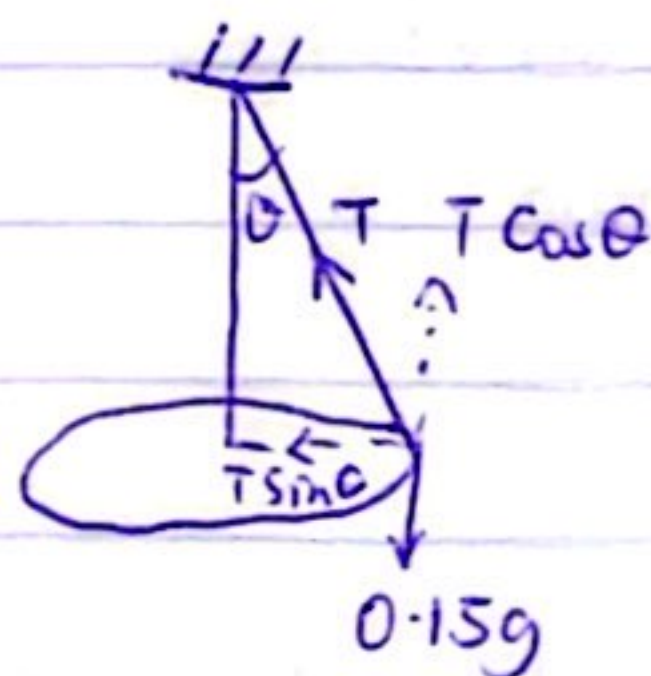
- Increasing strength of metals

- Improves durability

- Used in manufacturing eg rolling improves material performance.

(03 marks)

2. i)



$$v = \omega r$$

$$= \frac{10 \times 2\pi}{18} \times 1.82$$

$$= 6.3530 \text{ m s}^{-1}$$

(03 marks)

ii) $F = \frac{m v^2}{r}$

$$= \frac{0.15 \times (6.3530)^2}{1.82}$$

$$= 3.3264 \text{ N}$$

(03 marks)

iii) $\tan \theta = \frac{v^2}{r g}$

$$\tan \theta = (6.3530)^2 \div (1.82 \times 9.81)$$

$$\theta = 66.14^\circ$$

(03 marks)

iii)

$$T = m \omega^2 L$$

$$= 0.15 \left(\frac{10 \times 2\pi}{18} \right)^2 \times \left(\frac{1.82}{\sin 66.14} \right)$$

$$= 3.6373 \text{ N}$$

(03 marks)

END